

SHASHWAT MAHAJAN

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EDUCATION

Yeshwantrao Chavan College of Engineering, Nagpur

Expected June 2027

B.Tech in Computer Science Engineering | CGPA: 8.00 / 10 | Coursework: Machine Learning, DSA, Database Systems

EXPERIENCE

Generative AI Engineer Intern | Technical Hub | Generative AI Division

Jan 2026 – Present

- Designed and shipped LangChain-based RAG pipelines serving live end users; optimised vector retrieval by iterating on chunking strategy, embedding models, and re-ranking — improving response relevance in production.
- Applied systematic prompt evaluation and failure analysis to reduce hallucination rates; established monitoring baselines for ongoing retrieval quality tracking across multiple GenAI product initiatives.

PROJECTS

JobIntel Agent | Agentic AI Job Intelligence System [GITHUB](#)

Python · CrewAI · LangChain · Playwright · SendGrid · LinkedIn/Indeed/Naukri APIs · Vector Embeddings

- Architected a multi-agent CrewAI pipeline of four specialised agents — Scout, Analyst, Verifier, and Reporter — each with a defined role: sourcing jobs across LinkedIn, Indeed, Naukri, and company career sites; scoring fit; validating posting recency and legitimacy; and ranking results by priority.
- Implemented intent-based job matching using semantic vector embeddings over role descriptions and JD text, going beyond keyword search to capture contextual alignment between candidate profile and job requirements.
- Automated end-to-end intelligence delivery via SendGrid email report with ranked job listings, relevance reasoning, and direct application links — turning hours of manual search into a single scheduled pipeline run.

Financial Analyst RAG System | Production Research Syst [GITHUB](#)

Python · LangChain · ChromaDB · HuggingFace · RAGAS · AWS S3 · GitHub Actions

- Built a production RAG pipeline for real-time multi-company financial analysis; applied the RAGAS evaluation framework to benchmark retrieval faithfulness and answer relevance — enabling data-driven iteration on retrieval quality.
- Engineered HuggingFace query-expansion for cross-company comparison accuracy; implemented LLM output guardrails to constrain responses to grounded, cited content and reduce hallucination in financial contexts.

Credit Risk Assessment System | Full-Stack ML [GITHUB](#)

React.js · FastAPI · XGBoost · Docker · GitHub Actions · SMOTE

- Investigated severe class imbalance in a 10,000-record credit dataset using SMOTE oversampling; trained XGBoost classifier achieving 88% accuracy — meaningful improvement over ~70% majority-class baseline.
- Architected full inference pipeline (React.js → FastAPI → XGBoost) containerised with Docker and CI/CD via GitHub Actions, enabling reproducible zero-downtime deployments as a research-to-production artefact.

AWS Bedrock AgentCore Agent | Cloud-Native Agentic AI [GITHUB](#)

Python · LangGraph · AWS Bedrock · AWS Bedrock AgentCore · AgentCore Memory · DockerFastAPI · XGBoost · Docker · GitHub Actions · SMOTE

- Built and deployed a stateful AI agent using AWS Bedrock AgentCore and LangGraph.XGBoost classifier achieving 88% accuracy — meaningful improvement over ~70% majority-class baseline.
- Implemented AgentCore Memory for persistent memory, session continuity, and context retrieval.XGBoost classifier achieving 88% accuracy — meaningful improvement over ~70% majority-class baseline.
- Designed stateful agent orchestration workflows with long-term memory management and contextual reasoning.XGBoost classifier achieving 88% accuracy — meaningful improvement over ~70% majority-class baseline.
- Containerized runtimes with Docker and deployed cloud-native agent services on AWS infrastructure.XGBoost classifier achieving 88% accuracy — meaningful improvement over ~70% majority-class baseline.

HACKATHONS

DevOnboard — AI Repository Intelligence Platform | IBM Hackathon | Individual

Python · LangChain · RAG · Vector Embeddings · GitHub API

- Individually built an AI platform that ingests any GitHub repo via URL, parses its structure and source code, and returns cited context-aware answers to developer onboarding questions — compressing ramp-up from hours to seconds.
- Engineered a chunking-and-embedding retrieval pipeline with semantic vector search for accurate Q&A across large multi-file codebases with full source attribution; presented live demo to IBM panel.

Smart Attendance System | Smart India Hackathon (SIH) | Team Lead, 6-Member Team [GITHUB](#)

MongoDB · Express.js · React.js · Node.js · Supabase · PostgreSQL · WebSockets

- Led architecture design, sprint planning, and task delegation for a 6-member team; architected a real-time WebSocket layer validated under 100+ concurrent users with zero data loss.
- Designed rolling QR-based authentication to prevent replay attacks; implemented MFA with Supabase Row-Level Security for strict per-user data isolation.

SITNOVATE — Real-Time Chat Engine | 24-Hour Hackathon | Individual

React · WebSocket · JWT

- Scoped and delivered a JWT-secured real-time chat system entirely within 24 hours; made rapid architecture trade-offs under time constraints and shipped a live demo to a judging panel.
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TECHNICAL SKILLS

Generative AI & LLMs: LLMs, RAG, LangChain, LangGraph, CrewAI, AWS Bedrock, AWS Bedrock AgentCore, AgentCore Memory, Prompt Engineering, RAGAS, HuggingFace, Vector Embeddings, ChromaDB

Cloud & MLOps: AWS (Bedrock, AgentCore, S3), Docker, GitHub Actions, CI/CD

Languages: Python (primary), Java, JavaScript (ES6+), SQL, C

Systems & Web: FastAPI, Docker, GitHub Actions, AWS S3, React.js, Node.js, PostgreSQL, MongoDB, REST APIs, WebSockets

ACHIEVEMENTS & CERTIFICATIONS

LeetCode Rating 1538 — 400+ problems solved (Python & Java) | CodeChef 2-Star Coder

Machine Learning Specialization — Codebasics.io | Full Stack Web Development (MERN Stack)